

大学生创新创业训练计划项目结题报 告

面向极端热环境的高温红外屏蔽复合材料设计与性能优
化

项目组

2026 年 4 月 20 日

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第一章 基本信息

项目名称：面向极端热环境的高温红外屏蔽复合材料设计与性能优化

项目编号：XJ202511078412

项目周期：2025 年 03 月 – 2026 年 04 月

项目级别：校级

1.1 项目成员与分工

表 1.1: 项目成员信息表

项目角色	姓名	学号	所在院系	项目分工
负责人	裴开诚	32319600082	材料科学与工程学院	资金获取, 形式分析, 数据整理
成员 1	欧海轩	32319600008	物理与材料科学学院	资金获取, 形式分析, 数据整理
成员 2	刘霄昂	32319600071	材料科学与工程学院	资金获取, 形式分析, 数据整理
成员 3	欧杨轩	32407800096e	机械与电气工程学院	资金获取, 形式分析, 数据整理
第一导师	邢未未（讲师）	/	物理与材料科学学院	资金获取, 形式分析, 数据整理
第二导师	杜斌（副教授）	/	物理与材料科学学院	撰写-评审与编辑, 撰写-初稿, 资源支持, 形式分析

1.2 成果形式

学术论文一篇：doi: 10.1016/j.ceramint.2025.12.224

第二章 项目概述

2.1 研究背景与意义

随着微波通信技术的快速发展,电磁污染日益严重,不仅干扰电子设备,也对人体健康构成威胁。微波吸收材料(MAMs)被视为一种有效的解决方案,它们能将微波能量转化为其他形式,例如热能。目前已探索了多种 MAMs,包括一维材料(如 SiC 纳米线、碳纳米管等)、二维材料(如石墨烯、MXene 等)、以及块体材料(如碳基复合材料、陶瓷等)。然而,许多 MAMs 仍存在有效吸收带宽窄、密度过大的问题,限制了其满足实际应用中“薄厚度、低密度、强吸收、宽频带”的微波吸收需求。

在这些候选材料中,聚合物衍生陶瓷(PDCs)因其内在优势,近年来受到了广泛关注,包括分子水平的前驱体均一性、相较于传统陶瓷路线相对较低的加工温度,以及设计具有定制功能的新型成分的可行性。然而,传统的聚合物衍生陶瓷通常表现出有限的吸收带宽和较差的阻抗匹配,这导致入射微波被显著反射,最终限制了它们的实际应用。

最近的研究表明,对聚合物衍生陶瓷进行成分调控可以有效调节电磁参数并增强微波吸收。例如,Li 等人^[7]通过聚合物衍生陶瓷法制备了含有 Fe_3Si 球和 SiO_2 纳米线的 SiFeBOC 陶瓷,在仅 1.6 mm 厚度下实现了 $RL_{\min} = -47.68$ dB 的最小反射损耗和 4.39 GHz 的有效吸收带宽,这归功于磁电耦合、界面极化以及改善的阻抗匹配的协同效应。Chen 等人^[7]在 Sm 掺杂 SiBCN 陶瓷中提出了一种可控的相变策略,通过波透 Si_3N_4 、有损 SiC 和石墨的联合效应,实现了 $RL_{\min} = -54.24$ dB 和 7.48 GHz 的宽带宽。

2.2 国内外研究现状

近年来,国际上围绕硼化物、碳化物与氧化物复合体系开展了大量研究,重点关注高温结构稳定性与辐射特性耦合调控^[7-9]。Zhao 等^[1]通过在碳化硅材料表面镀银,形成导电网络通路,提高其电磁损耗;增强界面极化,提高其衰减常数和阻抗匹配,从而增强其吸波性能。Liu 等^[2]通过超快高温烧结的方式制备了三种新型 HEP 材料,通过调整组分并触发多种电磁损耗机制,该体系的电磁吸收性能得到了显著提升。Zhou 等^[3]通过化学气相渗透法构建了 $\text{SiC}_f/\text{Si}_3\text{N}_4$ 复合材料,实现了在宽温度范围 (25-600°C) 内 X 和 Ku 波段对温度不敏感的电磁波吸收。Lan 和 Wang 等^[4]通过在碳化硅晶须组成的轻质多孔骨架中掺入碳纳米线,热还原以实现碳化硅纳米线网络以产生强界面极化和多重反射,提高电导率。Cai 等^[5]通过在 1500 摄氏度、氮气气氛下烧结碳化硅晶须的方式实现了碳化硅/氮化硅复合气凝胶,将 Si_3N_4 纳米带引入碳化硅纳米线网络,增强了阻抗匹配、界面极化和偶极极化。我们总结了学界的部分代表性工作,见表 2.1,以供参考。

Material	Tickness (nm)	Minimum RL(dB)	EAB(GHz)	Ref.
Fe/SiC/PCS	2.25	-46.3	4.6(<-20dB)	[6]
MXene	2.7/2.1	-54.1(2.7)	7.76(2.1)	[7]
Fe/Co@C-CNFs	1.08/1.22	-18.66(1.08)	4.2(1.22)	[8]
PAN/Fe ₃ O ₄	4	-11.3	7	[9]
TiN/carbon-paraffin composites	1.9	-41.8	3.9	[10]
SiC	1.9	-57.8	12	[10]
ZnO/Co	3.0/2.6	-68.4(3.0)	5.9(2.6)	[11]
P-CNF/Fe	4.1	-44.86	3.28	[12]
CeO ₂ /NC	2.5	-42.95	8.48	[13]
Zr/SiC	3.5	-48.6	3.2	[14]
ZrC	1.25/1.0	-25.77(1.25)	3.04(1.0)	[15]
ZrO ₂ /ZrC/ZrB ₂	2.4	-21	8.64	[16]
ZrO ₂ /ZrB ₂ /C	4.0	-54	3.1	[17]
ZS-HNFA-x	2.4	-53.2	6.4	[18]
ZrO ₂ /CF-rGO	3.14	-62.99	8.19	[19]
ZrO ₂ /C	2.5	-36	7.3	[20] (本项目)

表 2.1: 不同高温红外屏蔽/微波吸收相关材料性能对比

2.3 研究目标

本项目立项时的核心目标如下：旨在开发一种具有非晶微观结构的新型 SiZrNOC 纳米纤维膜（NFM），以解决传统陶瓷纤维在高温环境下隔热性能不足的问题。团队通过：

- 引入 Zr 元素；
- 优化纤维的微观结构，提升其热稳定性和机械性能。

从而满足航空航天等领域对高性能隔热材料的需求。

第三章 研究内容与方法

3.1 主要研究内容

1. 高温红外屏蔽复合材料的配方设计与理论筛选；
2. 样品制备工艺参数优化（原料比例、烧结温度、保温时间）；
3. 高温红外性能、抗氧化性能与微观结构表征；
4. 建立性能指标与微结构参数的关联分析模型。

3.2 技术路线与方法

项目采用“理论筛选-实验制备-高温测试-机制分析-反馈优化”的闭环路线：

1. 通过文献与热力学判据初筛候选体系；
2. 制备不同组分与工艺参数样品；
3. 在目标温区测试 3–5 μm 与 8–14 μm 波段红外参数；
4. 结合 XRD、SEM 等结果分析相组成与界面结构；
5. 回代优化组分，形成稳定工艺方案。

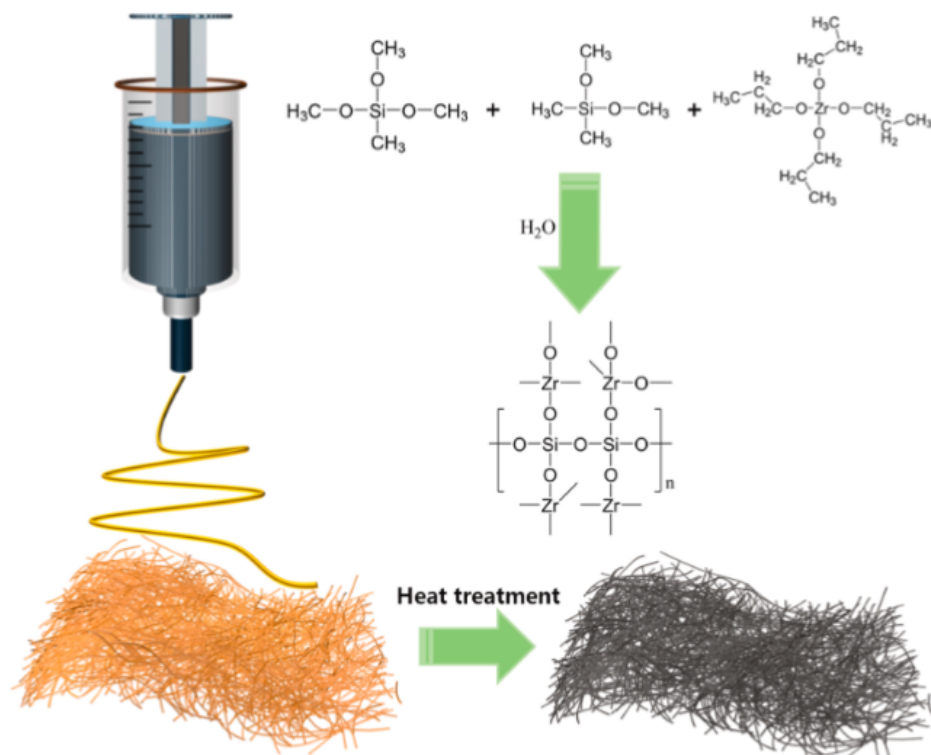


图 3.1: 项目技术路线图

3.3 实验条件

- 仪器设备：高温箱式炉、傅里叶红外光谱仪、XRD、SEM；
- 软件环境：Origin、MATLAB、jupyter（数据拟合与统计分析）；
- 样本来源：项目组自主制备样品。

第四章 项目执行情况

4.1 计划执行情况

总体执行与申报计划基本一致。

4.2 实施过程

- 第一阶段（2025.03-2025.05）：文献调研、方案设计、候选体系筛选；
- 第二阶段（2025.05-2025.07）：样品制备、参数摸索、初步性能测试；
- 第三阶段（2025.9-2026.12）：数据整合、机制分析、论文撰写、结题材料整理。

4.3 成员分工

负责人负责项目进度、实验统筹与结题文书；成员 1 负责样品制备流程与工艺复现；成员 2 负责红外与结构表征、数据处理与图表绘制；指导教师负责研究方向把控、论文框架修订与质量审核。组内采用每周例会机制进行任务同步与问题闭环。

第五章 研究成果与分析（核心章节）

5.1 主要研究结果

项目获得了具有代表性的高温红外屏蔽复合样品。与对照组相比，优化样品在目标波段表现出更低的平均发射率，并在高温氧化后仍保持较高结构稳定性。实验表明，适度提高致密相比比例并构建多尺度界面，可有效抑制高温辐射增强效应。

相关文献对比可见表 2.1。

表 5.1: 关键性能对比表:

组别	8–14 μm 平均发射率 (900 °C)	900 °C 氧化后质量保持率 (%)	抗热震循环次数 (次)
实验组	0.41	96.8	25
对照组	0.57	91.2	14

5.2 学术论文成果（重点）

论文信息:

- 标题: 面向极端热环境的高温红外屏蔽复合材料设计与性能优化
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- 作者排序: 项目负责人 (第一作者), 指导教师 (通讯作者)
- 项目资助标注: 本研究由广东省自然科学基金 (2023A1515030106)、国家自然科学基金 (52272058)、广州市科技计划项目基础与应用基础研究专题 (2023A04J1006)、广州市教育局材料科学与工程重点学科 (202255464)、广州大学本科生创新创业训练计划 (XJ202511078412) 以及广州大学研究生创新训练计划 (JCCX2025054) 提供资助。

论文核心内容提炼：论文围绕“高温稳定性与红外屏蔽性能协同提升”这一核心问题，构建了复合材料组分设计与工艺优化策略。通过对不同配比样品在中高温条件下的红外发射率、氧化增重与微观相演变进行系统表征，揭示了界面致密化程度与辐射特性之间的关联机制。研究结果表明，优化后的多相复合结构可在 900 °C 附近维持较低波段发射率，同时显著减缓氧化导致的性能衰退。论文进一步提出了面向工程应用的参数选取区间，为高温热管理和红外抑制材料开发提供了可复用的实验路径与数据依据。

第六章 创新点与特色

1. **面向高温服役场景的指标协同设计**：区别于单纯追求低发射率的方法，本项目同步考察耐温、抗氧化与热震稳定性，实现了多指标并行优化。
2. **组分-结构-性能的关联分析路径**：通过微观结构与红外参数的耦合分析，明确了关键组分比例与屏蔽效果之间的响应规律。
3. **可迁移的实验流程**：形成了从候选体系筛选到高温评价的标准化流程，可扩展至其他高温功能涂层体系。

第七章 总结与展望

7.1 项目总结

本项目围绕高温红外屏蔽材料的设计、制备与性能评估完成了预定目标，形成了较完整的数据体系与结题文档。研究表明，通过合理组分调控与工艺优化，可在保持高温稳定性的同时实现红外发射率降低，为后续工程化验证提供了基础。

7.2 存在的问题与不足

- 样本数量仍有限，统计显著性有待进一步增强；
- 部分工艺参数的耦合影响尚未完全解耦；
- 长时服役条件（>100 h）下的性能保持数据还不充分。

7.3 未来展望

- 扩展至更高温区（1000°C 以上）进行连续测试；
- 引入机器学习辅助配方筛选，提高实验效率；
- 推进中试级样件制备与复杂工况验证，支撑工程应用转化。

第八章 经费使用情况

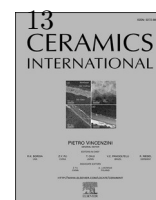
表 8.1: 经费执行情况表

科目	预算金额（元）	实际支出（元）	用途说明
材料费	980	980	化学试剂
测试费	1020	1020	校外测试：吸波性能测试费用
合计	2000	2000	执行率约 100%

第九章 附件与意见

9.1 附录文献全文

以下内容为所指定文献的完整 PDF 附录。



Electrospun SiZrOC nanofibers with tunable amorphous-crystalline structures for high-performance electromagnetic absorption

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Keywords:

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ABSTRACT

In polymer-derived ceramics, the electronegativity difference between Si and C dictates the covalent bonding topology and electron density distribution, thereby affecting polarization, electron transport, and electromagnetic (EM) wave attenuation. However, the role of amorphous-crystal heterostructures formed by in situ nanocrystal growth on polarization and EM absorption remains poorly understood. Here, SiZrOC nanofibers were fabricated via electrospinning followed by controlled pyrolysis, resulting in ZrO₂ nanocrystal embedded in an amorphous matrix. By modulating the pyrolysis temperature, the crystallization behavior of ZrO₂ nanocrystal and the corresponding dielectric properties were optimized, enabling balanced conductive loss, polarization loss, and multiple scattering. The nanofibers pyrolyzed at 1000 °C delivered exceptionally EM absorption properties, achieving a minimum reflection loss of −78.77 dB at 12.56 GHz with a thickness of 3.16 mm and a maximum effective absorption bandwidth of 6.8 GHz at a thickness of 3.88 mm. These results highlight the potential of SiZrOC nanofibers as lightweight, high-performance absorbers for next-generation wireless communication and high-power electronic devices.

1. Introduction

With the rapid advancement of microwave communication technologies, electromagnetic pollution has become increasingly severe, causing interference to electronic devices and posing risks to human health [1–4]. Microwave absorbing materials (MAMs) are considered an effective solution, as they convert microwave energy into other forms such as thermal energy [5–8]. Various MAMs have been explored, including one-dimensional materials (e.g., SiC nanowires, carbon nanotubes et al.) [9–12], two-dimensional materials (e.g., graphene, MXene et al.) [13–15], and bulk materials (e.g., carbon-based composites, ceramics et al.) [16–18]. However, many MAMs still suffer from narrow effective absorption bandwidths and excessive density, limiting their ability to meet the practical requirements of “thin thickness, low density, strong absorption, wide frequency band” absorption of microwave demands [19,20].

Among these candidates, polymer-derived ceramics (PDCs) have recently attracted considerable attention due to intrinsic advantages,

including precursor homogeneity at the molecular level, relative lower processing temperatures compared to conventional ceramic routes, and the feasibility of designing novel compositions with tailored functionalities [21,22]. Nevertheless, conventional PDCs generally exhibit limited absorption bandwidths and poor impedance matching, which results in significant reflection of incident microwaves and ultimately restrict their practical application [23,24]. Recent studies have demonstrated that compositional regulation of PDCs can effectively tune EM parameters and enhanced microwave absorption [25–27]. For example, Li et al. [28] fabricate SiFeBOC ceramics containing Fe₃Si spheres and SiO₂ nanowires via the PDC method, achieving a minimum reflection loss (RL_{min}) of −47.68 dB and an effective absorption bandwidth of 4.39 GHz at only 1.6 mm thickness, owing to the synergistic effects of magnetic-dielectric coupling, interfacial polarization, and improved impedance matching. Chen et al. [29] proposed a controllable phase-transformation strategy in Sm-doped SiBCN ceramics, achieving an RL_{min} of −54.24 dB and a broad bandwidth of 7.48 GHz through the combined effects of wave-transparent Si₃N₄ lossy SiC, and graphite

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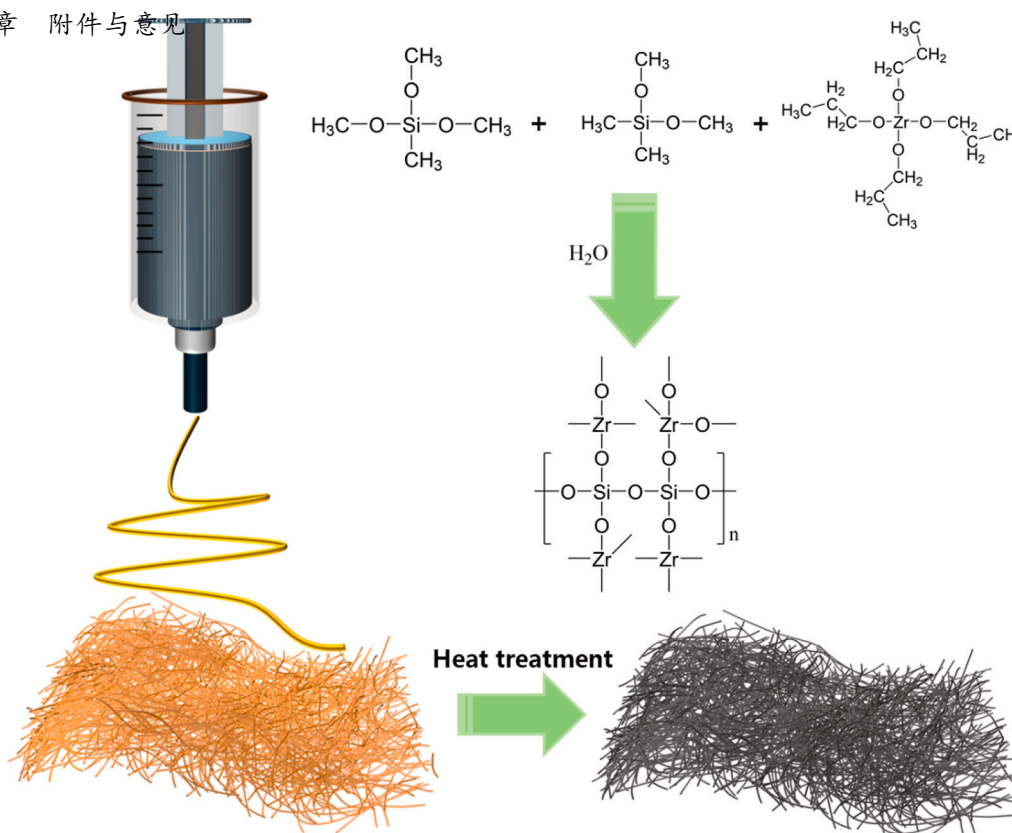


Fig. 1. Schematic illustration of the fabrication process of amorphous SiZrOC nanofibers.

strips. These studies highlight that compositional regulation can effectively balance impedance matching and dielectric loss, offering a promising path toward high-performance absorbers.

Besides composition tailoring, structural engineering plays a vital role in modulating EM responses [30–32]. Hierarchical structures such as porous frameworks, hollow spheres, core-shell architectures, and nanofibers have been shown to improve microwave attenuation by promoting multiple scattering and interfacial polarization [33–37]. Among these, electrospinning provides a versatile and scalable technique to fabricate continuous one-dimensional nanofibers with high aspect ratios, tunable pore porosity, and large surface-to-volume ratios [38–40]. These features enhance microwave transmission pathway and impedance matching, make electrospun nanofibers promising candidates for lightweight, high-performance MAMs. For example, Jiang et al. [41] fabricated a lightweight SiBCN/SiC nanowires aerogel, achieving an $RL_{\min}$ of -50.1 dB and a bandwidth of 6.4 GHz at just 2 mm thickness, attributed to improved impedance matching and multi-reflection effects. In a subsequent study, Jiang et al. [42] fabricated SiBCN/SiCN nanofiber ceramic aerogel using electrospinning, reaching an $RL_{\min}$ of -54.24 dB and a bandwidth of 7.48 GHz, further demonstrating the potential of electrospun structures. In our previous work [43] reported SiOC nanospheres fabricated by the PDCs route with excellent microwave absorption properties ($RL_{\min}$ of -56.82 dB at 2.93 mm thickness). However, limited effects have been made to further enhance the absorption of SiOC ceramics by combining the introduction regulation (secondary phase introduction) with structure design (nanofibers).

In this work, one-dimensional SiZrOC nanofibers were fabricated by electrospinning of silazirconoxane precursors and followed by controlled pyrolysis under argon. The electrospun nanofibers proved continuous transmission pathway, while the in-situ formation of ZrO_2 nanoparticles within the amorphous matrix introduces interfacial polarization and multiple scattering centers. As a results, the SiZrOC nanofibers exhibit excellent EM wave absorbing performance,

highlighting potential for applications in advanced communications systems and extreme environments.

2. Experimental methods

2.1. Materials

Methyltrimethoxysilane ($\text{CH}_3\text{Si}(\text{CH}_3\text{O})_3$), dimethyldimethoxysilane ($(\text{CH}_3)_2\text{Si}(\text{CH}_3\text{O})_2$), polyvinyl pyrrolidone (PVP), N,N-dimethylformamide (DMF), and zirconium acetylacetonate ($\text{Zr}(\text{acac})_4$) were purchased from Shanghai McLean Biochemical Technology Co., Ltd. All chemicals were used as received without further purification.

2.2. Preparation of amorphous SiZrOC nanofibers

Amorphous SiZrOC nanofibers were fabricated via electrospinning followed by pyrolysis at different temperatures. The precursor solution was obtained by homogeneously mixing methyltrimethoxysilane, dimethyldimethoxysilane, $\text{Zr}(\text{acac})_4$, PVP, DMF, H_2O at a weight ratio of $5:1:3:1:1$ under magnetic stirring at room temperature for 8 h. The resulting solution was electrospun using a spinning apparatus (E02, Foshan Weimai Technology Co., Ltd., China) onto an oil-coated paper collector (Fig. 1). The electrospinning parameters were set as follows: tip-to-collector distance, 15 cm; applied voltage, 20 kV; and feeding rate, 0.3 ml/h. The ambient temperature and relative humidity were maintained at $\sim 28^\circ\text{C}$ and $\sim 60\%$, respectively. After electrospinning, the obtained white precursor nanofibers were cured in air at room temperature for 24 h. Subsequently, they were pyrolyzed at different temperatures (800 – 1000°C) in argon atmosphere for 1 h at a heating rate of $2^\circ\text{C}/\text{min}$, yielding amorphous SiZrOC nanofibers.

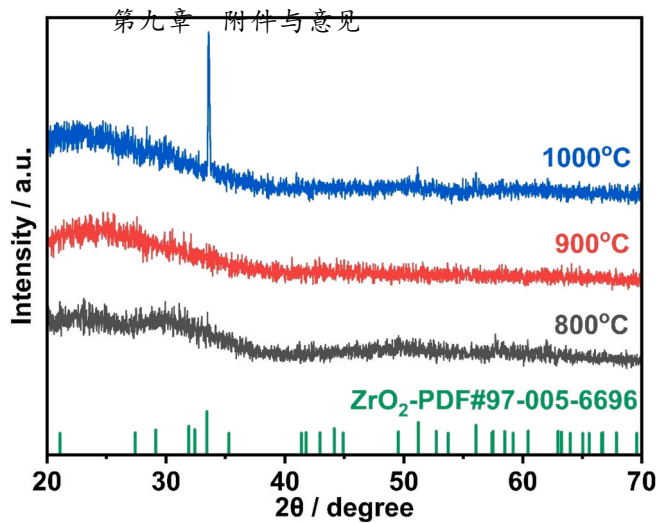


Fig. 2. XRD patterns of SiZrOC nanofibers pyrolyzed at different temperatures.

2.3. Characterization

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The phase composition was analyzed by X-ray diffraction (XRD, Ultima IV, Rigaku, Japan) using Cu K α radiation over 2θ range of 5–90°. The morphologies and microstructures were examined by scanning electron microscopy (SEM, JSM-IT800, JEOL, Japan) and transmission electron microscopy (TEM, CM300, Philips, Netherlands), both equipped with energy-dispersive X-ray spectroscopy (EDS). Elemental compositions and chemical states were determined by X-ray photoelectron spectroscopy (XPS, K-Alpha, Thermo Scientific, USA).

The EM properties were measured using the transmission/reflection (T/R) coaxial line method on a vector network analyzer (N5230A, Agilent, USA) in the frequency range of 2–18 GHz. For measurement, SiZrOC nanofibers were uniformly mixed with paraffin (30–50 wt% powder content) and pressed into cylindrical rings with an inner diameter of 3.04 mm, outer diameter of 7.00 mm, and thickness of 2.0–4.0 mm.

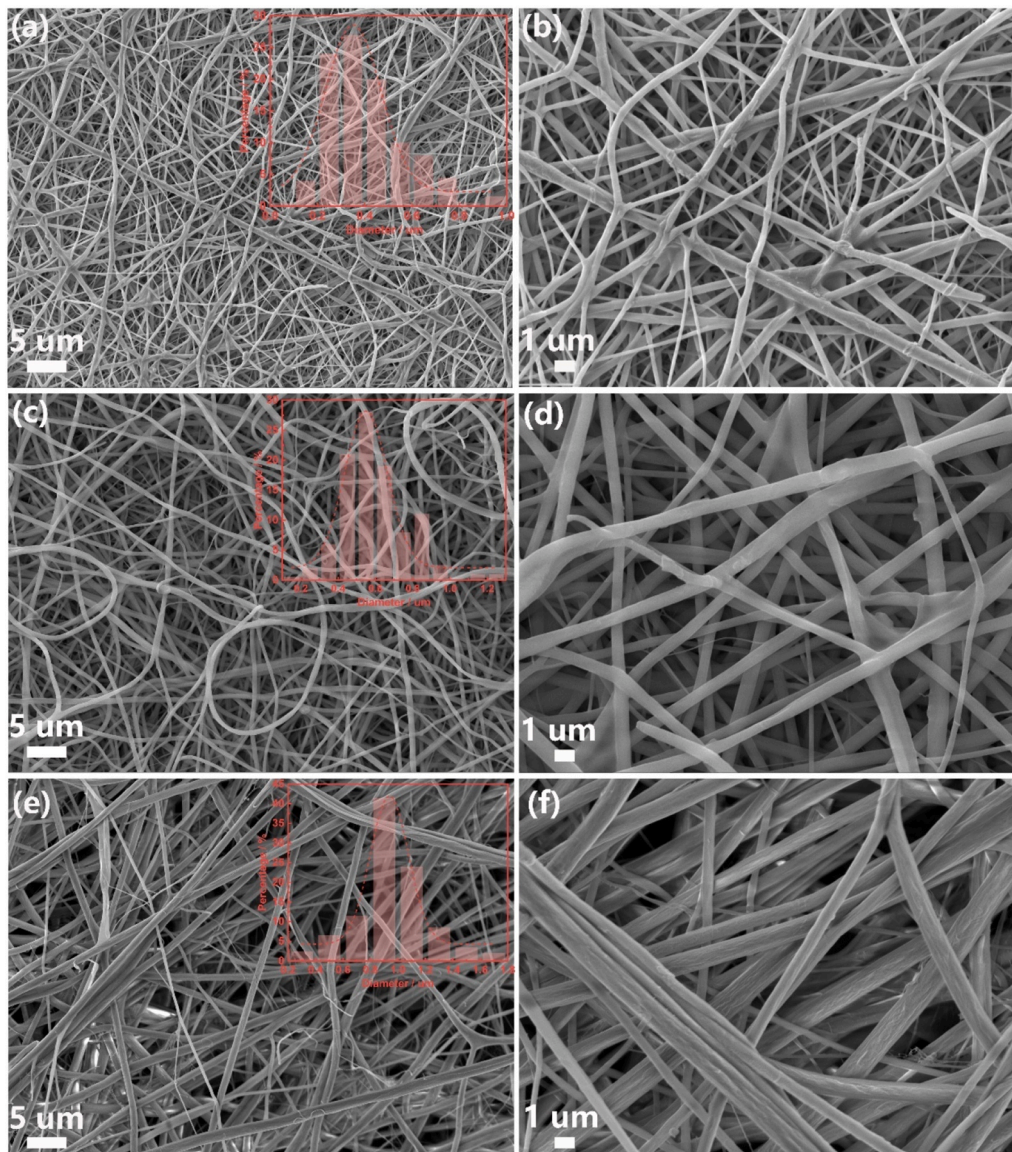


Fig. 3. SEM images of amorphous SiZrOC nanofibers pyrolyzed at different temperatures (a, b) 800 °C; (c, d) 900 °C; (e, f) 1000 °C.

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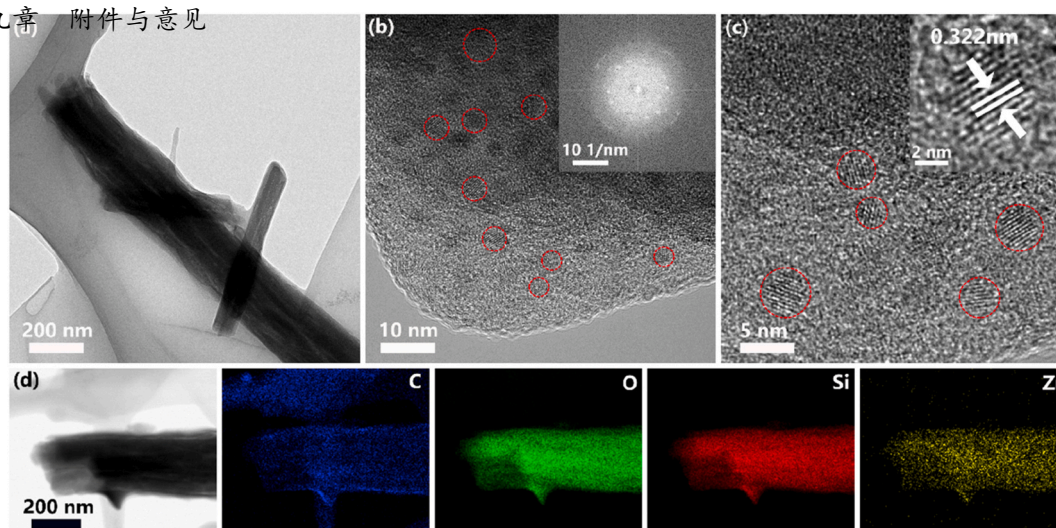


Fig. 4. Microstructure of the amorphous SiZrOC nanofibers heat-treated at 1000 °C: (a) TEM image; (b, c) HRTEM images and SAED pattern: nanocrystals within the amorphous matrix; (d) elemental mappings of Si, Zr, O, and C.

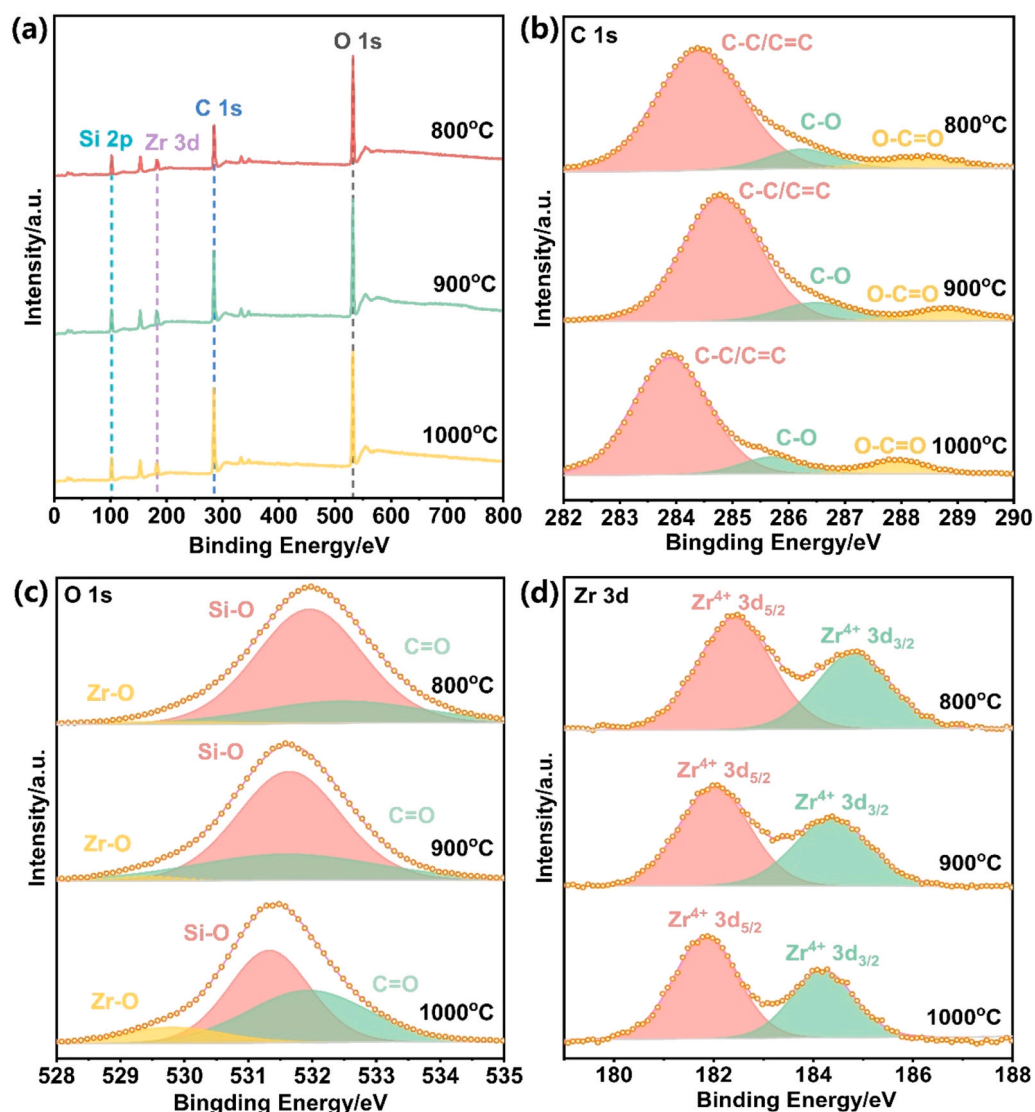


Fig. 5. XPS spectra of SiZrOC nanofibers: (a) wide-scan XPS spectra; high-resolution spectra of (b) C1s, (c) O1s, and (d) Zr 3d.

3. Results and discussion

3.1. Phase and morphology of the amorphous SiZrOC nanofibers

Fig. 2 shows the XRD patterns of SiZrOC nanofibers pyrolyzed at different temperatures. At lower pyrolysis temperatures, the broad diffraction hump suggests a predominantly amorphous structure, typical of PDCs at lower pyrolysis temperatures. With increasing pyrolysis temperature, distinct diffraction peaks gradually emerge, corresponding to the crystallization of phase, ZrO_2 . When the temperature reaches 1000 °C, a sharp peak appears at $\sim 34^\circ$, accompanied by additional weaker peaks at around 28° and 50° , which can be indexed to the ZrO_2 phase. These results, consistent with previous reports [39], suggest that phase separation occurs during pyrolysis, leading to the gradual formation of ZrO_2 within the amorphous structure.

Fig. 3 illustrates representative SEM images of SiZrOC nanofibers pyrolyzed at different temperatures, highlighting their temperature-dependent morphological evolution. At 800 °C the fibers show an average diameter of approximately 360 nm and retain a uniform amorphous morphology characterized by smooth surface and a well-interconnected network. When the pyrolysis temperature is increased to 900 °C, the average fiber diameter expands to about 550 nm. At this stage, partial fiber fusion and a noticeable increase in surface roughness emerge, which can be attributed to thermally activated structural rearrangement and the onset of localized sintering. With further temperature elevation (e.g., ≥ 1000 °C), the fiber diameter increases to approximately 940 nm. This coarsening is accompanied by more pronounced surface roughening, consistent with crystallization and phase-separation phenomena commonly observed in polymer-derived ceramic systems. These progressive structural transformations strongly influence the microstructural integrity and, consequently, the functional performance of the nanofibers.

3.2. Microstructure and chemical characteristic of the amorphous SiZrOC nanofibers

TEM analysis was carried out to further confirm the structural characteristics of the amorphous SiZrOC nanofibers. As shown in Fig. 4a, the fibers exhibit diameters of approximately 200 nm, with some thinner fibers measuring around 100 nm, in good agreement with the SEM observations. High-resolution TEM (HRTEM) reveals numerous nanocrystalline domains (Fig. 4b and c) embedded within the amorphous matrix, forming abundant heterogeneous interfaces. The SAED pattern (Fig. 4b) exhibits a dominant amorphous ring due to the bulk amorphous SiOC matrix, but also contains weak and diffuse diffraction spots originating from the nanometer-scale ZrO_2 crystallites. Such coexistence—an amorphous halo with superimposed faint nanocrystal reflections—is characteristic of PDC undergoing early-stage phase separation and nanocrystallization. The lattice fringes observed in Fig. 4c show a measured interplanar spacing of ~ 0.322 nm, which is consistent with the (020) planes spacing of orthorhombic ZrO_2 . Elemental mappings show the uniform distribution of C, O, Si, and Zr, validating the successful fabrication of SiZrOC nanofibers. This hybrid nanostructure, comprising nanocrystals embedded in an amorphous matrix, is advantageous for EM wave attenuation. On one hand, the amorphous matrix contains a high density of defects of structural defects; on the other hand, the abundant heterogeneous interfaces between the nanocrystals and the matrix promote interface polarization. These synergistic factors effectively promote dielectric loss.

The chemical bonding states of the SiZrOC nanofibers were investigated by XPS (Fig. 5). The wide-scan XPS spectra (Fig. 5a) confirms the presence of Si, Zr, O, and C elements in fibers pyrolyzed at 800 °C, 900 °C, and 1000 °C. High-resolution spectra of C1s, O1s, and Zr3d provide further insights into the evolution of chemical states with pyrolysis temperature. The C1s spectrum (Fig. 5b) displays a major peak at ~ 284.6 eV, attributed to C-C/C=C bonds (graphitic carbon). Additional

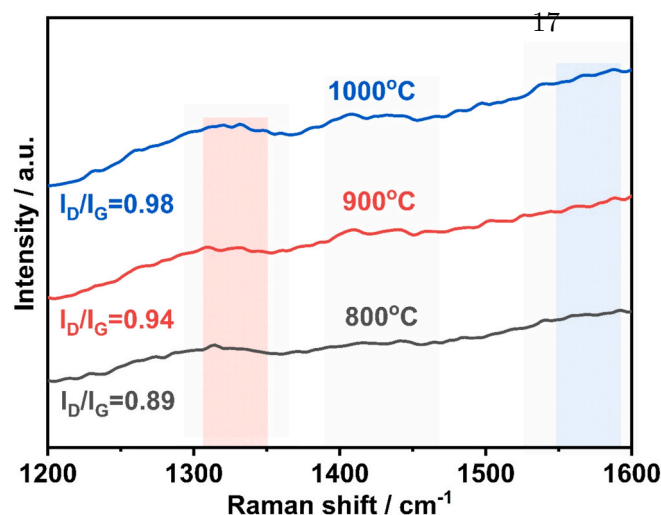


Fig. 6. Raman spectra of SiZrOC nanofibers pyrolyzed at different temperatures.

peaks at around 286.2 eV and 288.5 eV correspond to C-O and O-C=O species, respectively. With increasing pyrolysis temperature, the intensity of oxygenated carbon peaks decreases, indicating the decomposition of organic residues and purification of the carbon phase. The O1s spectra (Fig. 5c) shows two peaks at ~ 531.3 eV, associated to Si-O and Zr-O bonds, respectively, along with a weaker peak at around 532.4 eV, corresponding to C=O groups. Increasing pyrolysis temperature leads to a stronger inorganic oxygen signal and weaker high-binding-energy components, suggesting the growth of oxide networks. The Zr3d spectra (Fig. 5d) exhibit two peaks at ~ 182.3 eV and ~ 184.7 eV, corresponding to Zr 3d_{5/2} and Zr 3d_{3/2} of Zr^{4+} in ZrO_2 . The gradual enhancement and slight shift of these peaks toward higher binding energies with increasing temperature reflect progressive oxidation and structural ordering of the Zr-based network. Collectively, these results indicate that increasing the pyrolysis temperature drives the transformation from an organic-rich precursor to a hybrid nanostructure composite of Zr-O inorganic frameworks and partially graphitized carbon. This structural evolution provides synergistic regulation of conductive, dielectric, and interfacial components, resulting in optimized microwave absorption. Raman spectra further confirm the evolution of the carbon phase. As can be seen from Fig. 6, when the temperature increases from 800 °C to 1000 °C, two prominent peaks gradually appear approximately 1330 cm^{-1} and 1570 cm^{-1} for the SiZrOC fibers, corresponding to the disordered (D) and graphitization (G) peaks, respectively. The intensities of both D and G peaks increase with pyrolysis temperature, which signifies the gradual transformation of amorphous carbon into ordered graphitic domains. The I_D/I_G values for 800 °C, 900 °C, and 1000 °C are 0.89, 0.94, and 0.98, respectively. The gradual increase in I_D/I_G ratio with temperature indicates the presence of more defect structures and increased disorder with emerging graphitic ordering in the SiZrOC fiber matrix pyrolyzed at 1000 °C. Such defect-rich carbon structures are known to act as effective polarization centers, facilitating dipole polarization, hopping conduction, and defect-induced relaxation process.

3.3. EM wave absorbing performance of the SiZrOC nanofibers

The complex permittivity ($\epsilon = \epsilon' - j\epsilon''$) of SiZrOC nanofibers was measured to evaluate their EM response over the 2–18 GHz frequency range. Fig. 7 presents the real (ϵ') and imaginary (ϵ'') parts of permittivity for nanofibers pyrolyzed at 800 °C (a and b), 900 °C (c and d), and 1000 °C (e and f) with varying fiber contents (30 wt%, 40 wt%, and 50 wt% in paraffin). At 800 °C, the nanofibers show relatively high ϵ' and ϵ'' values, particularly for the 50 wt% samples, with $\epsilon' > 6$ and $\epsilon'' > 2.0$

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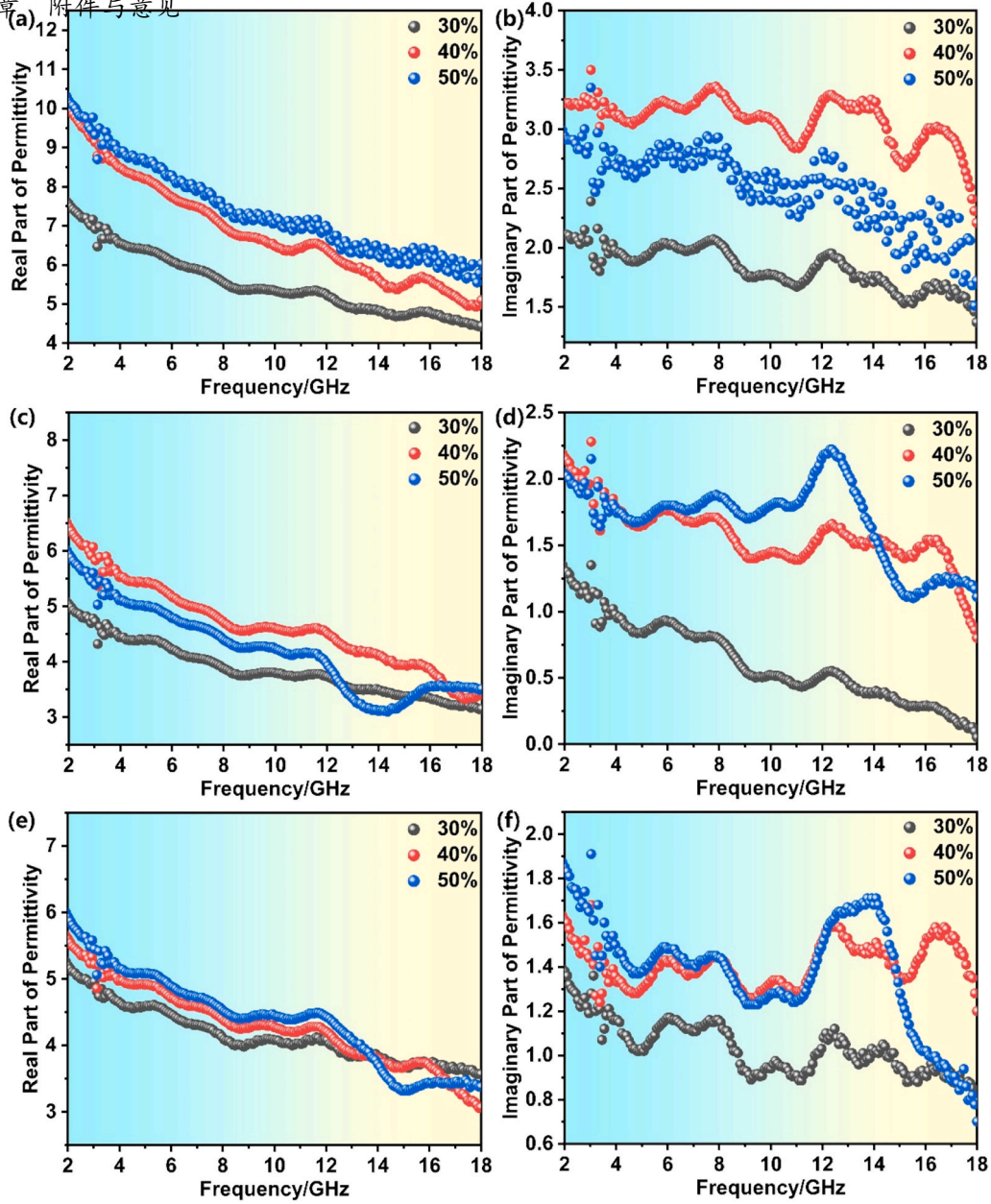


Fig. 7. EM properties of SiZrOC nanofiber at different pyrolysis temperature and fiber contents: (a, b) 800 °C, (c, d) 900 °C, and (e, f) 1000 °C; (a, c, e) real part of permittivity (ϵ'), (b, d, f) imaginary part of permittivity (ϵ'').

across much of the frequency range, indicating strong polarization capability and stronger dielectric loss. Both ϵ' and ϵ'' gradually decrease with increasing frequency, attributed to weakened dipole orientation polarization. For nanofibers pyrolyzed at 900 °C, ϵ' and ϵ'' decrease slightly compared to the 800 °C sample, indicating that partial crystallization or phase transformation begins to reduce the complex permittivity. Notably, the ϵ'' values of 50 wt% samples remain above 1.5 across most the Ku-band, suggesting sustained dielectric loss. At 1000 °C, both ϵ' and ϵ'' reach their lowest values among all samples, with ϵ' dropping below 5 for the 30 wt% fibers and ϵ'' generally below 1.5. This decrease is attributed to extensive formation of ZrO_2 phases, which possess relatively low dielectric constants. At a fixed pyrolysis temperature, increasing fiber content from 30 to 50 wt% enhances both ϵ' and ϵ'' values, likely due to the formation of more continuous conductive or semi-conductive networks within the paraffin matrix.

Fig. 8 shows the dielectric loss angular tangent ($\tan \delta = \epsilon'' / \epsilon'$), reflecting the material's ability to dissipate EM energy. The 50 wt%

sample pyrolyzed at 1000 °C exhibits significantly higher $\tan \delta$ compared to other lower-temperature samples (Fig. 8e), resulting from amorphous phases and defects structures that promote interfacial and dipolar relaxation. For fiber pyrolyzed at 800 °C and 900 °C, $\tan \delta$ fluctuates between 0.2 and 0.6 (Fig. 8a and c), indicating enhanced conductivity due to higher degree of graphitization. At a fixed temperature, increasing fiber content generally increase $\tan \delta$, although an exception is observed for the 40 wt% sample at 800 °C. Given that the material is essentially non-magnetic, dielectric loss dominates the absorption mechanism. The dielectric response can be described by Debye's theory, where ϵ' and ϵ'' are described as follows [44–46]:

$$\epsilon' = \epsilon_{\infty} + \frac{\epsilon_s - \epsilon_{\infty}}{1 + (2\pi f \tau)^2} \quad (1)$$

$$\epsilon'' = \frac{2\pi f \tau (\epsilon_s - \epsilon_{\infty})}{1 + (2\pi f \tau)^2} + \frac{\sigma}{2\pi f \epsilon_0} \quad (2)$$

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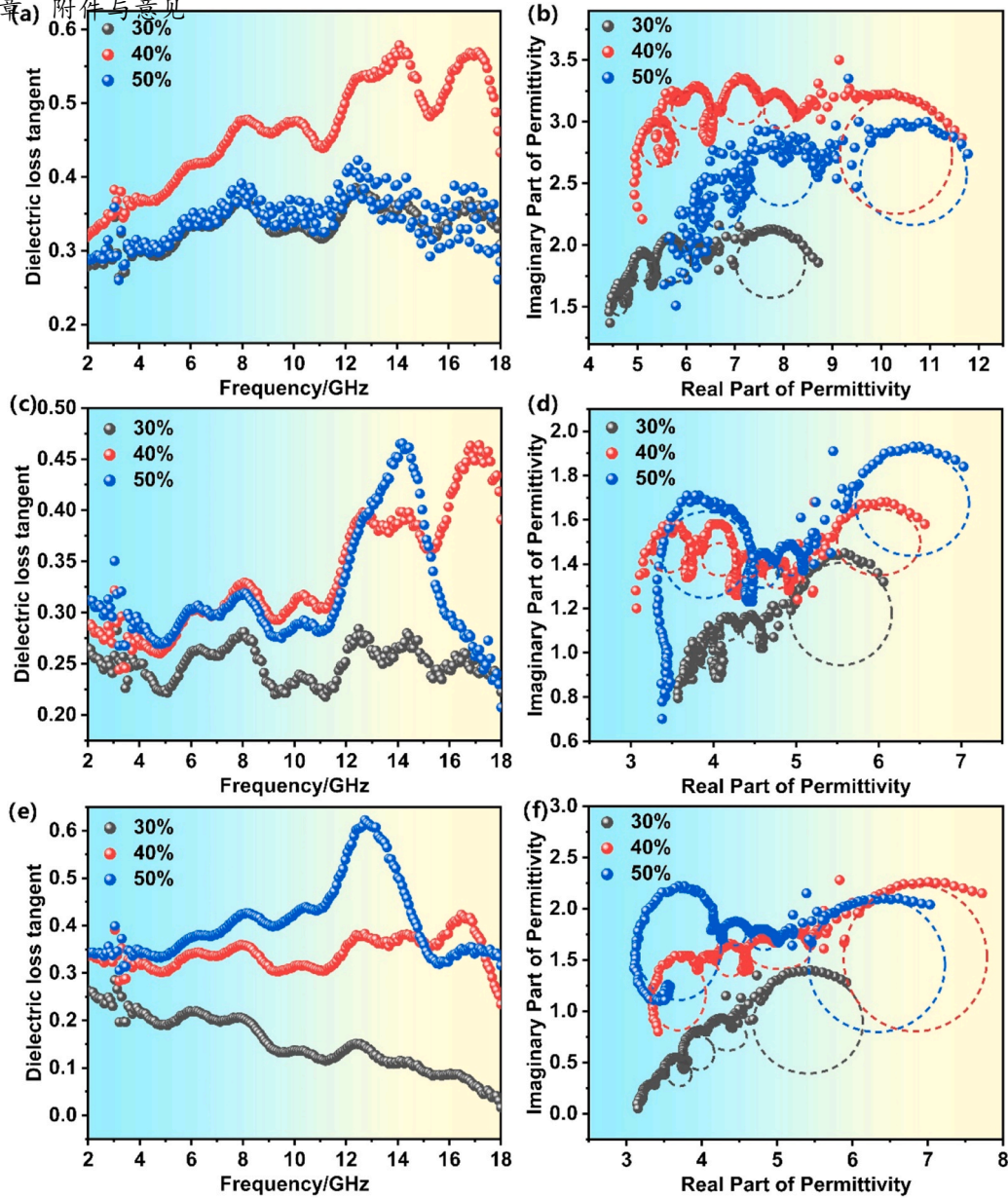


Fig. 8. Dielectric loss tangents and Cole-Cole plots of SiZrOC amorphous nanofibers pyrolyzed at different temperatures: (a, c, e) dielectric loss tangent ($\tan \delta$), (b, d, f) Cole-Cole semicircles; (a, b) 800 °C, (c, d) 900 °C, (e, f) 1000 °C.

where ε_{∞} and ε_s are the high-frequency and static permittivity, f is the frequency; τ is the relaxation time, and σ is the conductivity of the sample. In Equation (2), the first term represents the polarization relaxation, while the second term accounts for the conduction. The relationship between of ε' and ε'' can be further described as [47–49]:

$$\left(\varepsilon' - \frac{\varepsilon_s + \varepsilon_{\infty}}{2}\right)^2 + (\varepsilon'')^2 = \left(\frac{\varepsilon_s - \varepsilon_{\infty}}{2}\right)^2 \quad (3)$$

Equation (3) produces semicircular ε' - ε'' curves, known as Cole-Cole plots, representing Debye relaxation process. Fig. 8b, d, and f show the Cole-Cole plots of SiZrOC nanofibers. Multiple semicircles are observed for all the samples, indicating the coexistence of multiple polarizations mechanisms contributing to relaxation loss. The tails at low frequencies confirm the presence of conduction loss. Higher pyrolysis temperatures enhance both conductive loss and polarization loss, attributed to increasing graphitization (Ohmic loss) and interfacial polarization at defect sites and ZrO₂ interface. Therefore, the dielectric and conduction

loss dominate the EM wave absorption of the amorphous SiZrOC nanofibers.

According to transmission line theory, the EM wave absorption of SiZrOC nanofibers can be evaluated by reflection loss (RL), which is calculated from the EM parameters as follows [50,51]:

$$RL(dB) = 20 \log \left| \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \right| \quad (4)$$

$$Z_{in} = Z_0 \sqrt{\frac{\mu_r}{\varepsilon_r}} \tanh \left[j \left(\frac{2\pi f d}{c} \right) \sqrt{\mu_r \varepsilon_r} \right] \quad (5)$$

where Z_{in} and Z_0 are the input impedance of the material and the free space impedance, respectively; d is the absorber thickness and c is the speed of light in vacuum. Usually, $RL < 10$ dB suggest that over 90 % of the incident EM wave is absorbed, and the corresponding frequency range is defined as the effective absorption bandwidth (EAB). Fig. 9 displays the RL of SiZrOC nanofibers pyrolyzed at different temperatures

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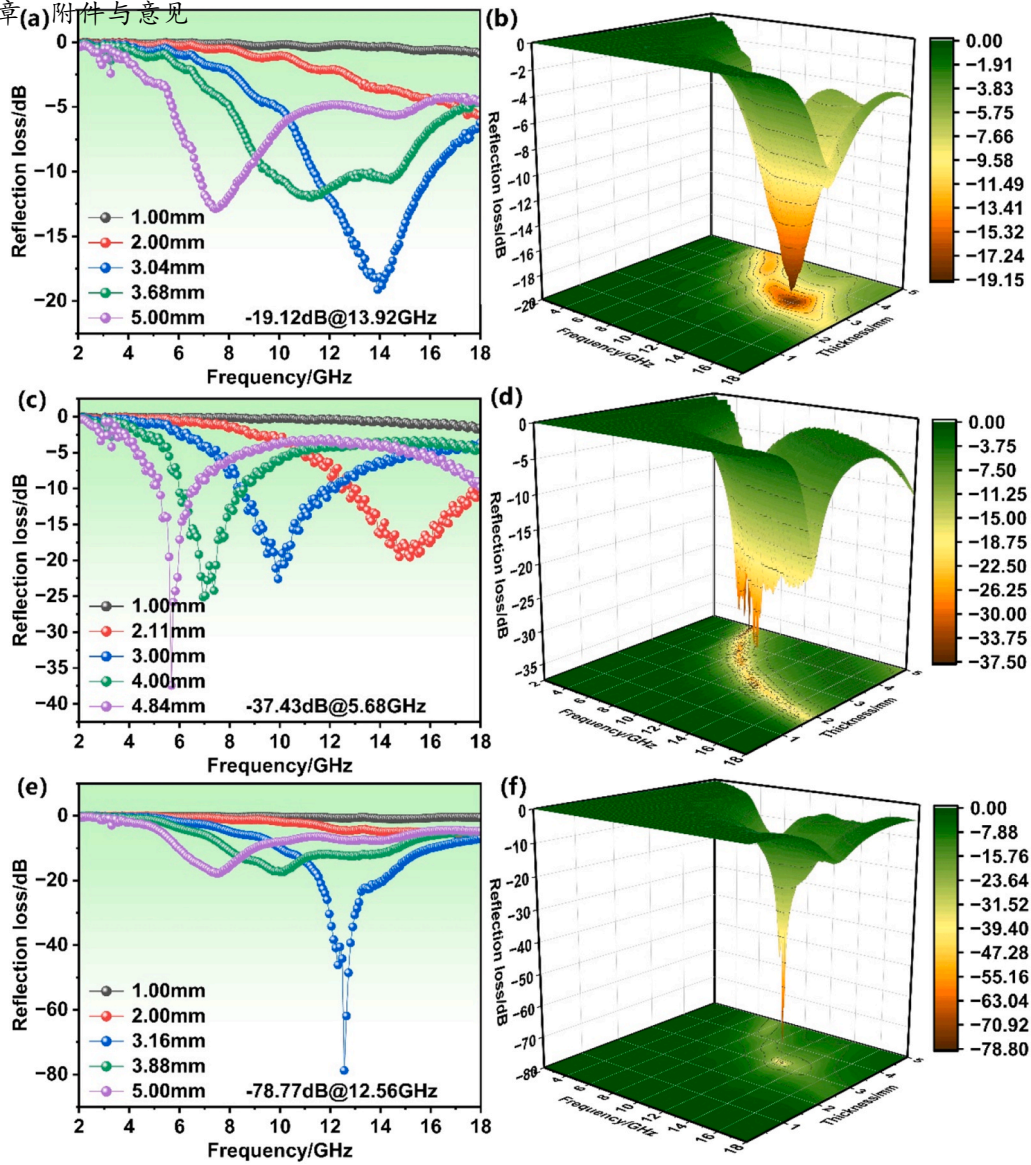


Fig. 9. The two-dimensional and three-dimensional reflection loss (RL) of SiZrOC nanofibers (50 wt% content) as a function of frequency and thickness: (a, b) pyrolysis at 800 °C; (c, d) pyrolysis at 900 °C; (e, f) pyrolysis at 1000 °C.

with 50 wt% fiber content. The others show in Figs. S1–S2 (Supporting information). As depicted in Fig. 9e and f, the sample pyrolyzed at 1000 °C exhibits an impressive RL_{min} of -78.77 dB at 12.56 GHz with a thickness of 3.16 mm and an EAB_{max} was 6.8 GHz at a thickness of 3.88 mm, revealing its optimal EM wave absorption performance. This excellent performance is attributed to numerous interfaces between the amorphous matrix and ZrO_2 nanocrystals, which enhance interfacial polarization and dielectric loss. By comparison, the samples pyrolyzed at 800 °C and 900 °C show RL_{min} value of -19.12 dB at 13.92 GHz and -37.43 dB at 5.68 GHz (Fig. 9a–d), respectively, which are higher due to fewer interfaces and reduced interfacial polarization. Notably, the 900 °C sample with 40 wt% fiber (Fig. S2, supporting information) content also exhibits excellent absorption, with RL_{min} value of -60.23 dB at 11.54 GHz with the thickness of 2.62 mm, and an EAB_{max} value of 5.2 GHz at the thickness of 6.89 mm (Table S1, supporting information). These results confirm that EM absorption strongly dependent on the pyrolysis temperatures, as higher temperature promote the formation of more nanocrystals, increasing interfacial polarization, relaxation polarization loss, and multiple scattering.

Dielectric properties are the primary factor in EM absorption, but

impedance matching ($Z = Z_{in}/Z_0$) and attenuation constant (α) are also crucial. Impedance matching determines whether EM waves can effectively enter the material rather than being reflected at the surface. Ideal impedance matching occurs when $Z \approx 1$, meaning the material impedance equals free space impedance. The attenuation coefficient α determines the ability of the material to dissipate EM waves that have entered, and is calculated as [52–54]:

$$\alpha = \frac{2\pi f}{c} \times \sqrt{(\mu''\epsilon'' - \mu'\epsilon') + \sqrt{(\mu''\epsilon'' - \mu'\epsilon')^2 + (\mu''\epsilon'' + \mu'\epsilon')^2}} \quad (6)$$

Fig. 10a–c displays the impedance matching maps for 50 wt% SiZrOC nanofibers at different pyrolysis temperatures. In the 4 – 10 GHz range, a thickness of 3 – 5 mm allows the Z to approach 1 , while in the 10 – 18 GHz range, a thickness of 1 – 3 mm provides ideal matching. The 1000 °C sample achieves $RL_{min} = -78.77$ dB at 12.56 GHz with a thickness of 3.16 mm, coinciding with the best impedance matching. Fig. 10d shows the attenuation constant α , which increase with frequency for all samples, indicating strong EM wave attenuation. The 1000 °C sample reaches a maximum α of 176.55 around 12 GHz, demonstrating that the combination of excellent impedance matching and high α enables

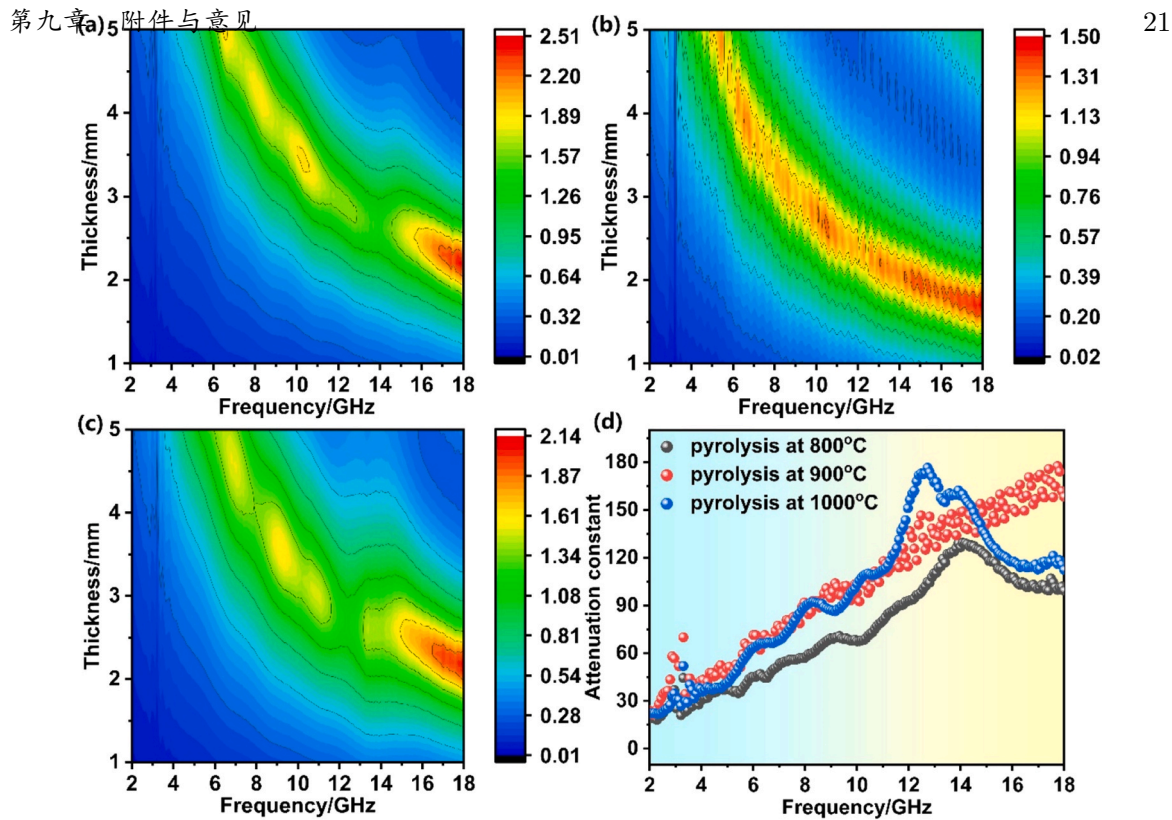


Fig. 10. Impedance matching maps and attenuation constant (α) of SiZrOC nanofibers (50 wt% content) pyrolyzed at different temperatures: (a–c) impedance matching maps for 800 °C, 900 °C, and 1000 °C, respectively; (d) attenuation constant (α).

superior absorption performance.

3.4. Microwave absorption mechanism of the SiZrOC nanofibers

The dielectric loss of different samples was fitted using the least squares method, as shown in Fig. 11a and b. The conduction loss is inversely proportional to frequency and directly proportional to conductivity (Fig. 11a). Meanwhile, higher pyrolysis temperatures enhance the conduction loss of SiZrOC nanofibers. From Fig. 11b, it can be observed that the polarization loss of SiZrOC nanofibers decreases with increasing pyrolysis temperature. This is because as the temperature rises, the microstructure coarsens, oxygen-containing functional groups gradually decompose, and ZrO₂ nanocrystals precipitate. These changes reduce the number of effective dipoles and interfacial polarization centers. At the same time, the increasing graphitization degree of the free carbon phase enhances electrical conduction, causing part of the dielectric loss mechanism to shift from polarization loss to conductive loss. The results indicate that conduction loss and polarization loss dominate the EM wave attenuation of the 1000 °C pyrolyzed sample, primarily due to its unique structure. The main mechanisms of EM wave absorption in SiZrOC nanofibers (50 wt%) pyrolyzed at 1000 °C can be understood from both microscopic and macroscopic perspectives, as shown in Fig. 11c. At the microscopic level, defects such as vacancy, dangling bonds, and the residual active functional sites in the SiZrOC nanofibers act as polarization centers under the alternating electromagnetic field, resulting in dipole polarization loss and enhanced electromagnetic field, resulting in dipolar polarization loss and enhanced absorption. Additionally, in the SiZrOC nanofibers (50 wt%) pyrolyzed at 1000 °C, the amorphous matrix and numerous in-situ formed ZrO₂ nanoparticles generated abundant heterogeneous interfaces. These interfaces facilitate charge accumulation, leading to strong space charge polarization and increased interfacial loss. Higher pyrolysis temperatures promote the formation of free carbon, enhancing the electrical

conductivity of the material. The increased conductivity facilitates carrier migration and hopping, inducing field-driven microcurrents and further enhancing conduction loss. From a macroscopic point of view, the highly porous structures of the SiZrOC nanofibers (50 wt%) pyrolyzed at 1000 °C provides multiple transmission pathways for incident EM wave, improves impedance matching, and promotes multiple reflections. Therefore, under the synergistic effects of interfacial polarization, dipole polarization, conduction loss, and multiple reflections, the SiZrOC nanofibers (50 wt%) pyrolyzed at 1000 °C exhibit outstanding microwave-absorbing properties.

The microwave absorption performance of SiZrOC nanofibers compared with other representative materials prepared via electrospinning are summarized in Table 1. The results clearly demonstrate that SiZrOC nanofibers exhibit a significantly lower RL_{min} value and a relatively broader EAB_{max} , outperforming most reported electrospun counterparts. In summary, the SiZrOC nanofiber materials display excellent microwave absorption performance, highlighting their potential in the field of microwave absorption.

4. Conclusion

In this study, SiZrOC nanofibers were successfully fabricated via electrospinning followed by controlled pyrolysis. By tuning the pyrolysis temperature, both the microstructure and EM properties of the samples can be effectively optimized, leading to improved impedance matching and enhanced attenuation capability. The sample pyrolysis at 1000 °C, consisting primarily of ZrO₂ nanoparticles embedded in an amorphous matrix, exhibits exceptional EM wave absorption properties due to the synergistic effects of its unique microstructure, enhanced conductive loss and polarization loss, and the macroscopic porous architecture. Notably, this sample achieves a minimum reflection (RL) value of -78.77 dB at 12.56 GHz with a thickness of 3.16 mm and a maximum effective absorption bandwidth was 6.8 GHz at a thickness of 3.88 mm.

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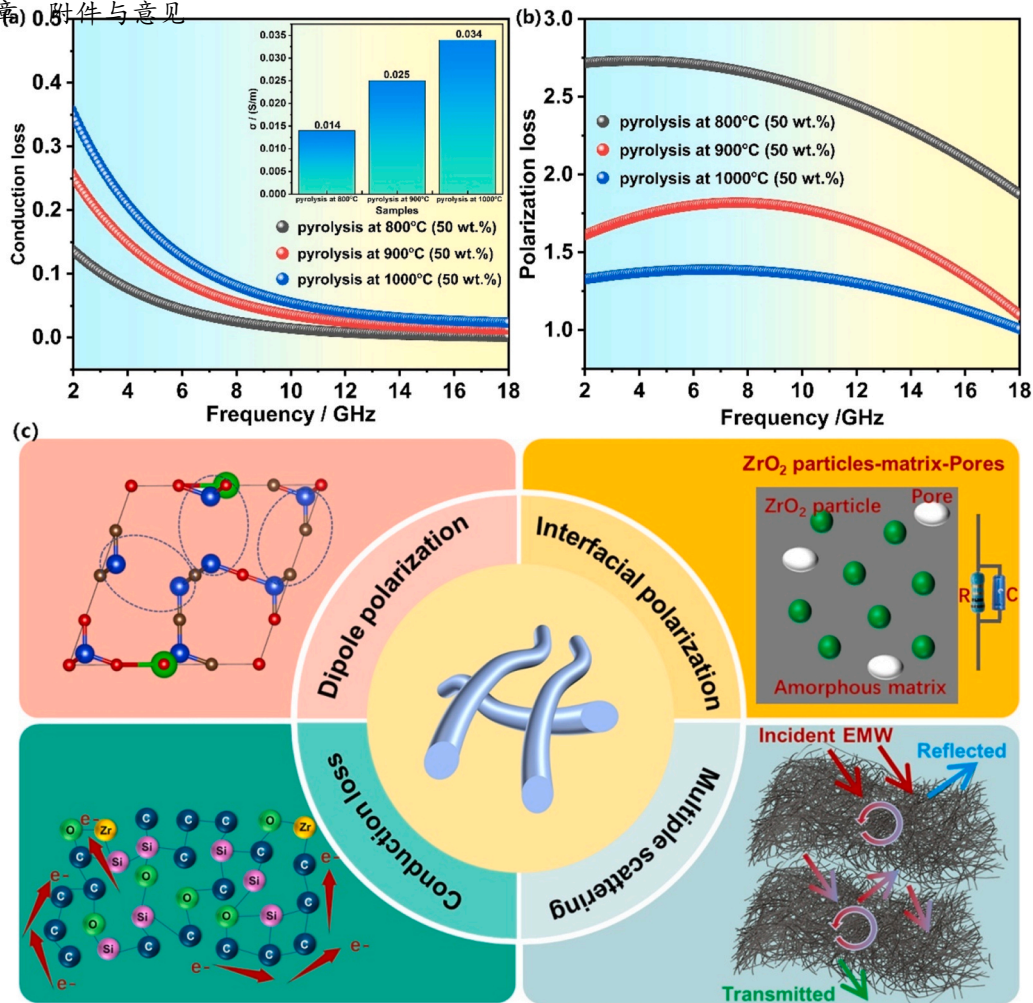


Fig. 11. Schematic illustrations of the EM wave absorption mechanisms of SiZrOC nanofibers (50 wt%) pyrolyzed at 1000 °C: (a) conduction loss; (b) polarization loss, and (c) overall absorption mechanism.

Table 1

Comparison of microwave-absorbing performance among SiZrOC nanofibers and other representative materials prepared via electrospinning.

Material	Thickness/ mm	RL _{min} / dB	EAB _{max} / GHz	Ref.
MFO/CGFS-3	4.8	-57.21	3.58	[2]
Fe _{0.64} Ni _{0.36} /MXene/ CNFs	2.7	-54.1	7.76	[4]
Fe ₃ O ₄ @NRC	1.5	-37.4	4.16	[5]
Fe/Co@C-CNFs	1.08	-18.66	4.2	[7]
SiC/C	3.0	-36	4.1	[9]
HCSS/Zr/SiC	3.5	-48.6	3.2	[11]
ZrO ₂ /CF-rGO	3.14	-62.99	8.19	[30]
ZS-HNFA-x	2.4	-53.2	6.4	[33]
ZrO ₂ /ZrB ₂ /C	4.0	-54	3.1	[35]
ZrO ₂ /ZrC/ZrB ₂	2.4	-21	8.64	[37]
SiCN	2.0	-31.1	4.59	[40]
CeO ₂ /N-C	2.5	-42.95	8.48	[55]
Ni PNFs	2.5	-40.4	4.6	[56]
SiZrOC	3.16	-78.77	6.8	This work

This work demonstrates, for the first time, the significant contribution of a crystal-amorphous hybrid structure to the dielectric loss in SiZrOC nanofibers and provides a valuable guideline for designing highly efficient electromagnetic wave absorbers.

CRediT authorship contribution statement

Saidi Wang: Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Haixuan Ou:** Funding acquisition, Formal analysis, Data curation. **Kaicheng Pei:** Funding acquisition, Formal analysis, Data curation. **Yangxuan Ou:** Funding acquisition, Formal analysis, Data curation. **Xiaoang Liu:** Funding acquisition, Formal analysis, Data curation. **Yimin Ouyang:** Funding acquisition, Formal analysis, Data curation. **Weiwei Xing:** Funding acquisition, Formal analysis, Data curation. **Bin Du:** Writing – review & editing, Writing – original draft, Resources, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ceramint.2025.12.224>.

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